

UNIT 8

TYPES OF PRINTER



Learning Objectives

1. Define printers as output devices and describe their purpose.
2. Differentiate between impact and non-impact printers.
3. Explain the working mechanism of various types of printers, including dot matrix, line, ink-jet, and laser printers.
4. Identify the advantages and limitations of each type of printer.
5. Perform a lab activity to explore the functionality of a printer.

Computers are powerful machines that process data, but to share the results of this data processing with users, they need **output** devices. **Output devices** are hardware components that convey information from the computer to the user. Common output devices include **monitors, speakers, and printers**. Among these, printers hold a significant place as they provide a permanent output of digital data on paper, such as documents, images, and more. This chapter focuses on printers and their various types.

Printers

Printers are output devices used to prepare permanent outputs on paper. They transform digital information into a physical format, making them essential for producing hard copies of text, images, or graphics.

Types of Printer

Printers can be divided into two main categories:

1. Impact Printers

Impact printers use an electro-mechanical mechanism, where hammers or pins strike against a ribbon and paper to print the text. These printers are further divided into:

a. Character Printer:

- Prints one character at a time.
- Operates at a slower speed compared to other printers.
- Example: Dot Matrix Printer.

Dot Matrix Printer:

- Prints characters as a combination of dots.
- The print head consists of a matrix of pins (usually 9 or 24) that strike a carbon ribbon to transfer the pattern onto the paper.
- Commonly used for printing receipts and invoices.



b. Line Printer:

- Prints an entire line of text at once.
- Was commonly used in the early days of digital computing and is still used in some industrial applications.



Non-Impact Printers

Non-impact printers use modern technologies like ink-jet or laser mechanisms, providing higher quality output at faster speeds. These printers are quieter and more efficient than impact printers. They are divided into:

a. Ink-Jet Printer:

- Sprays patterns of ink onto paper through tiny nozzles.
- Produces high-quality images and text.
- Commonly used in homes and offices.



b. Laser Printer:

- Utilizes a laser beam to create an image on a drum.
- The drum attracts toner, which is then transferred to paper using heat and pressure.
- Known for speed and precision, with speeds ranging from 4 to 20 pages per minute (ppm).



EXERCISES

01: Choose the correct answer.

1. What is the primary purpose of a printer?

- a) Inputting data into the computer
- b) Producing permanent output on paper
- c) Storing digital files
- d) Enhancing computer speed

2. Which category of printers uses an electro-mechanical mechanism to print?

- a) Non-impact printers
- b) Impact printers
- c) Ink-jet printers
- d) Laser printers

3. Which type of printer prints characters as a combination of dots?

- a) Line printer
- b) Ink-jet printer
- c) Dot matrix printer
- d) Laser printer

4. What distinguishes non-impact printers from impact printers?

- a) Non-impact printers use hammers or pins.
- b) Non-impact printers are quieter and faster.
- c) Non-impact printers print one character at a time.
- d) Non-impact printers use ribbons for printing.

5. Which of the following is an example of a character printer?

- a) Ink-jet printer
- b) Dot matrix printer
- c) Laser printer
- d) Line printer

6. How does an ink-jet printer create an image?

- a) By striking pins on a ribbon
- b) By spraying patterns of ink through nozzles
- c) By using a laser beam to transfer toner
- d) By printing one line of text at a time

7.What is a common use for a dot matrix printer?

- a) Printing high-quality images
- b) Printing receipts and invoices
- c) Printing large quantities of documents at high speed
- d) Printing photographs

8.Which printer uses a laser beam to create an image on a drum?

- a) Ink-jet printer
- b) Dot matrix printer
- c) Laser printer
- d) Line printer

9.What is a key advantage of a laser printer over an ink-jet printer?

- a) Lower quality output
- b) Higher speed and precision
- c) Slower printing process
- d) No use of toner

10.Which printer is most likely to be used in an industrial setting for printing large amounts of text?

- a) Ink-jet printer
- b) Dot matrix printer
- c) Line printer
- d) Laser printer

02: Fill in the blanks.

1. The impact printers are divided into _____ and _____.
2. The non - impact printers are divided into _____ and _____.
3. The _____ printer prints characters as a combination of dots.
4. _____ printers use non-impact technology for better quality output.
5. A _____ printer prints an entire line of text at once.
6. Laser printers use a _____ beam to produce an image on a drum.

03: Write True or False for each statement.

	TRUE	FALSE
Common input devices include monitors, speakers, and printers.	☐	☐
Common output devices include mouse and keyboard.	☐	☐
Dot Matrix Printers are an example of non-impact printers.	☐	☐
Ink-Jet Printers spray ink onto paper to create text and images.	☐	☐
Laser Printers are slower than Dot Matrix Printers.	☐	☐
Line printers are mostly associated with the early days of computing.	☐	☐

04: Answer the following questions.

1. Define a printer and its purpose.
2. Differentiate between impact and non-impact printers.
3. Explain the working of a dot matrix printer.
4. What makes laser printers faster and quieter than impact printers?

[illegible]

[illegible]

LAB ACTIVITIES

Lab Activity 1: Inserting and Formatting Images with Advanced Options



1. Objective: To explore the functionality of printers in the lab.

- Connect a computer to an available printer (e.g., ink-jet or laser printer).
- Print a test page to observe the quality and speed.
- Identify the type of printer and note its key features.

2. Tasks:

- Compare the test pages from an ink-jet printer and a laser printer.
- Discuss the differences in quality, speed, and noise levels.

3. Report: Write a short report summarizing your findings from the lab activity.